LAPORAN PRAKTIKUM INTERNET OF THINGS

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POLITEKNIK NEGERI LAMPUNG

JURUSAN TEKNOLOGI INFORMASI

PROGRAM STUDI TEKNOLOGI REKAYASA INTERNET

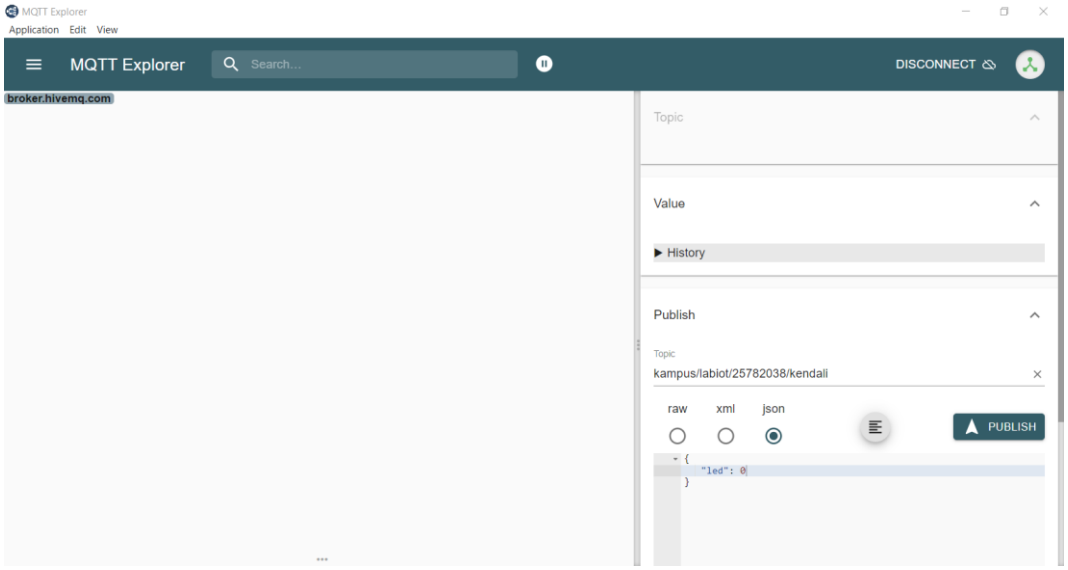
2026

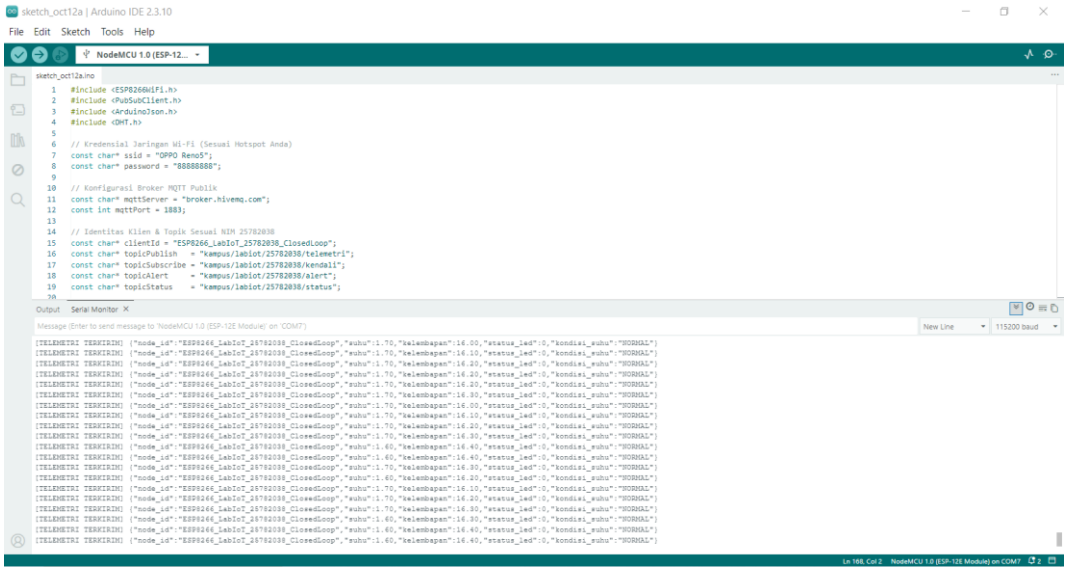
6.7 Tugas: Sistem Telemetri & Kendali Closed-Loop Industri

Tujuan Utama: Menggabungkan pembacaan sensor suhu dan kelembapan, pengemasan data JSON secara dinamis, kendali LED jarak jauh, kanal peringatan darurat terpisah, serta fitur keandalan *Last Will and Testament* (LWT).

- **Rule Engine & Safety Logic:**

  - **Klasifikasi Suhu:** Menambahkan kunci `"kondisi_suhu"` pada data JSON telemetri, di mana nilainya `"NORMAL"` jika suhu $\le 31.0^\circ\text{C}$ dan `"PANAS"` jika suhu $> 31.0^\circ\text{C}$.
  - **Kanal Alert Mandiri:** Jika suhu melebihi 31.0°C, sistem secara otomatis mempublikasikan peringatan ke topik khusus (`.../alert`) dengan status `BAHAYA` dan menyalakan LED indikator tanpa menunggu perintah operator. Sistem juga menggunakan pengaman (*state latch*) agar pesan peringatan tidak memenuhi *broker* secara terus-menerus di setiap milidetik.
  - **Last Will and Testament (LWT):** Mendaftarkan pesan wasiat pada topik status (`.../status`) agar jika mikrokontroler terputus secara mendadak (misalnya kabel dicabut), *broker* akan otomatis mengumumkan bahwa perangkat berstatus `OFFLINE`.
  - **Manual Override:** Saat suhu dalam kondisi aman (normal), operator luar tetap bisa menyalakan atau mematikan LED secara manual melalui topik kendali (`.../kendali`).


MQTT Explorer
Application Edit View
MQTT Explorer
Search...
DISCONNECT
broker.hivemq.com
Topic
Value
History
Publish
Topic
kampus/labiot/25782038/kendali
raw
xml
json
PUBLISH
{
"led": 0
}


sketch_oct12a | Arduino IDE 2.3.10
File Edit Sketch Tools Help
NodeMCU 1.0 (ESP-12...
sketch_oct12a.ino
#include <ESP8266WiFi.h>
#include <PubSubClient.h>
#include <ArduinoJson.h>
#include <DHT.h>
// Kredensial Jaringan Wi-Fi (Sesuai Hotspot Anda)
const char* ssid = "OPPO Reno5";
const char* password = "88888888";
// Konfigurasi Broker MQTT Publik
const char* mqttServer = "broker.hivemq.com";
const int mqttPort = 1883;
// Identitas Klien & Topik Sesuai NIM 25782038
const char* clientId = "ESP8266_LabIoT_25782038_ClosedLoop";
const char* topicPublish = "kampus/labiot/25782038/telemetri";
const char* topicSubscribe = "kampus/labiot/25782038/kendali";
const char* topicAlert = "kampus/labiot/25782038/alert";
const char* topicStatus = "kampus/labiot/25782038/status";
Output
Serial Monitor
Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM7')
New Line
115200 baud
[TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782038_ClosedLoop","suhu":1.70,"kelembapan":16.00,"status_led":0,"kondisi_suhu":"NORMAL"}
[TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782038_ClosedLoop","suhu":1.70,"kelembapan":16.10,"status_led":0,"kondisi_suhu":"NORMAL"}
[TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782038_ClosedLoop","suhu":1.70,"kelembapan":16.20,"status_led":0,"kondisi_suhu":"NORMAL"}
[TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782038_ClosedLoop","suhu":1.70,"kelembapan":16.20,"status_led":0,"kondisi_suhu":"NORMAL"}
[TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782038_ClosedLoop","suhu":1.70,"kelembapan":16.20,"status_led":0,"kondisi_suhu":"NORMAL"}
[TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782038_ClosedLoop","suhu":1.70,"kelembapan":16.30,"status_led":0,"kondisi_suhu":"NORMAL"}
[TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782038_ClosedLoop","suhu":1.70,"kelembapan":16.00,"status_led":0,"kondisi_suhu":"NORMAL"}
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[TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782038_ClosedLoop","suhu":1.70,"kelembapan":16.30,"status_led":0,"kondisi_suhu":"NORMAL"}
[TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782038_ClosedLoop","suhu":1.60,"kelembapan":16.40,"status_led":0,"kondisi_suhu":"NORMAL"}
[TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782038_ClosedLoop","suhu":1.60,"kelembapan":16.40,"status_led":0,"kondisi_suhu":"NORMAL"}
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[TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782038_ClosedLoop","suhu":1.60,"kelembapan":16.40,"status_led":0,"kondisi_suhu":"NORMAL"}
[TELEMETRI TERKIRIM] {"node_id":"ESP8266_LabIoT_25782038_ClosedLoop","suhu":1.60,"kelembapan":16.40,"status_led":0,"kondisi_suhu":"NORMAL"}
Ln 168, Col 2 NodeMCU 1.0 (ESP-12E Module) on COM7

